Results in Engineering 25 (2025) 104050

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Mobile Application to Locate Charging Stations for Electric Vehicles and highlight its specifications

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ARTICLE INFO

Keywords:
Electric Vehicles
Charging Stations
Mobile Application
Smart Mobility Infrastructure

ABSTRACT

Increased demand for EVs has created a need for a new infrastructure, especially in locating, accessing and efficiently utilizing a charging station based on individual user needs. This paper describes an EV charging station location & companion application designed for Android mobile devices in Java and XML in Android Studio, outlining a modular design for the application allowing it to operate on-line while demonstrating off-line (via a demo) works with the user. The demo prototype used data gathering and structured local JSON formatted input data sets to closely mimic the cloud API response via JSON that would be required for the real application to work online. The prototype was tested functionally across several configurations demonstrating rapid parsing via Gson of JSON data inputs, smooth operational performance using Google Maps on all device formats while maintaining an average of around 1.5 seconds response time for all data render cycles. The current research is a first step in working toward scalable smart mobility aligned with national missions towards EV adoption and a precursor to developing cloud, user faced charging station data services.

1. INTRODUCTION

Electric Vehicles (EVs) are becoming a household staple across the world, replacing fossil fuel powered automobiles due to the growing awareness about climate change and sustainable development among people, thanks to dedicated missions by the United Nations (UN). According to International Energy Agency (IEA), sale of electric cars exceeded 17 million globally in 2024, indicating more than 20% increase in sales compared to the previous year [1]. It is general knowledge at this point that EVs require proper charging setups to keep them running. While home charging ports are generally preferred due to their convenience, there is a dire need for ubiquitous setups of public charging stations to support mass adoption of EVs among the segments of population without access to home charging [1].

In 2024, over 1.3 million public charging points were added worldwide, a rise of more than 30% from the previous year. The additions made that year alone nearly matched the total global stock of 2020. Since 2020, approximately two-thirds of the expansion in public chargers has taken place in China, which now has around 65% of all charging points and 60% of the world's electric light-duty vehicles [1]. India has begun the EV revolution as well, with policy initiatives to advance charging accessibility. Around 40,000 new public chargers were installed in 2024, and in October that year, the government allocated INR 20 billion under the PM E-DRIVE scheme to expand charging infrastructure, prioritizing urban areas and major transport corridors. [1,2].

The early efforts on EV charging infrastructure have focused mainly on easing information flow, coordination, and direction to users, representing previous work done to develop technical dimensions of the application intended to ease the discovery of charging stations. Lee et al. [3] have developed a model for cooperation using blockchain technology in which charging stations rely on verified shared information while also using mobile charging units to alleviate congestion and reduce wait times. Later, Ibrahim et al. [4] developed a recommendation system that included charging speed and cost to make recommendations, reflecting an increased demand for decision support systems that fit the user-defined experience. Research on dynamic routing has also evolved in this manner. For example, Dastpak et al. [5] have included real-time occupancy information to make route optimizations in real-time. Feng et al. [6] similarly included sustainability-based criteria to recommend optimal station locational information for EV charging stations. Visaria et al. [7] and Touker et al. [8] reiterate that ease of use and access are significant influences on the adoption of EVs. Accordingly, all this work emphasizes the need for accurate and easy-to-read station location information and the benefit of condensing those details into a mobile application.

Meanwhile, a parallel line of research has been conducted on forecasting demand and predicting station occupancy. For example, Orzechowski et al. [9] developed a deep-learning model capable of predicting station demand at a medium-term forecast based on historical occupancy as well as predicting uptake of charging stations, meanwhile Wang et al.[10] and Ma and Faye [11] developed LSTM-based frameworks for short-term forecasting, as well as multi-step occupancy prediction, with high accuracy. To further enhance this, Cao et al. [12] included uncertainty for probabilistic demand forecasting.

Reinforcement learning can also be utilised in real-time real-time station recommendation, with Xu et al.'s [13] graphical based framework considering traffic and power limitations simultaneously. Routing models such as Zhang et al.'s (2020) BiS4EV [14], and Pourazarm et al.'s [15] dynamic programming approach use intelligent navigation to improve charging efficiency. Though these models are at a systems level, their insights emphasize the need for mobile applications as implementation considerations because they provide users with availability indicators, wait-time estimates, and alternative stations, which ultimately creates a more predictable and less stressful charging experience.

A wider body of work studies grid interaction, infrastructure planning, and all the consideration factors for improving the quality and consistency of data made available to the user-facing tools. A paper by Zhang et al. [16] outlined a bi-layer optimization model with the user navigation at one level and grid constraints on the second layer. Lee et al. [17] showed real world benefits of adaptive scheduling at the network scale. Reviews by Kunj and Pal [18] and Motlagh et al highlight IoT-based smart charging, V2G integration, and market-aware coordination [19]. Challenges of interoperability, as identified by van der Kam and Bekkers [20], further elaborate the need for applications to collect information across different networks. Blockchain-based frameworks [21][22], like the one proposed by Iqbal et al. and Motlagh and Li, address security and standardization requirements from a systems perspective, whilst issues around infrastructure, raised by Babic et al. and Mao et al.[23][24], reinforce the need for accurate demand predictions, and effective planning around placement.

These studies indicate that besides a growing research interest in optimizing systems, day-to-day EV users have no simple and digestible tools available to combine charging station locations and parameters, waiting times, and information about amenities close by.

This project seeks to approach the gap by enabling ease of locating charging stations and add other functionalities that helps the user with port specifications, expected wait times and information about other necessary locations (like landmarks) located around the charging stations.

The user simply indicates a target destination, and the application provides information about each charging station located along that journey, including access to information about restaurants/pharmacies nearby to wait while they charge. Further, server-side access will be given to the administrator of the charging stations management system to allow them to manage all appropriate information.

The project will engage real-time data architectures, geo-location intelligence, and interface optimization strategies in the android ecosystem. The application will support both:

1. Live Mode: Real-time data fetched using REST APIs from OpenChargeMap (the worlds largest open registry with updated information since 2011) and Google Places API.
2. Demo Mode: Carefully structured local JSON datasets (ev_features.json, ports.json, cars.json), simply structured to simulate responses from the API to have a full simulation within the front-end interface and maintain code integrity between testing and production environments.

The project demonstrates an extendable foundation for potentially cloud deployment and eventual linkage with Firebase Realtime Database or the server with a custom back end utilizing modular components, such as Gson for json object deserialization, Retrofit/Volley for handling APIs asynchronously, and cMaterial design-based UI cards for greater visual hierarchy.

2. CONTRIBUTION TO KNOWLEDGE:

The contributions of this research are as follows:

1. Dual-Mode - a unique hybrid architecture which can operate in both online and offline mode, all the while maintaining the same architecture.
2. Data Integration - a unified presentation of station metadata, charger compatibility matrices, and amenity map for context.
3. Performance - evidenced efficient parsing of JSON (mean latency of 1.5 seconds) and smooth UI rendering across a broad hardware profile.
4. Scalable architect - a modular architecture capable of upgrades, including machine learning recommendation engines, availability prediction models, and smart grid integrations.

The remainder of this document outlines the technical method, system architecture, integration logic, testing method, and potential ways to enhance this multi-layered mobile solution.

3. RESEARCH DESIGN AND DEVELOPMENT

The method of development is scalable and replicable, with context to deployment. This is consistent with current best practice for contemporary software engineering in developing mobile applications particularly in resource-limited and variable network conditions like the emerging EV markets. Each functional module could be validated independently and in both online (live API) and offline (local JSON) execution modes before ensuring consistent user experiences regardless of network conditions.

4. SYSTEM ARCHITECTURE DESIGN

The system was developed as a three-tier modular pipeline, separating concerns for presentation, application logic, and persistence/data layers:

1. Live Mode Architecture:
   - The application invokes REST API calls to live data providers to dynamically retrieve the charging station information. The charging station information comes from the following:
   - OpenChargeMap API: The OpenChargeMap API offers a global, comprehensive, and open database of electric vehicle charging station locations, along with metadata about those stations, including latitudes/longitude of the locations of the charging stations, types of chargers located at those sites, specifications around the kilowatt output of the charging stations (3.7 kW, 7 kW, 22 kW, 50 kW, 150 kW), whether the charging station is operational, and any other comments that a user may have entered at that site.
   - Google Places API: The Google Places API provides context to the charging station data by locating additional information regarding restaurants, cafes, rest areas, hospitals, and other retail stores in the vicinity of charging stations within a specific radius.
2. Demo Mode Architecture:
   - In order to circumvent the network dependency of the live mode and to enable full testing offline, the application provides local structured JSON datasets that replicate and mirror the schema and hierarchy of data provided by the live API, as follows:
   - ev_features.json: Provides structured arrays of amenity features indexed by lat/lon.
   - ports.json: A complete specification of supported charger port specification data with technical metadata described.
   - cars.json: A comprehensive database of electric vehicle types with associated compatibility matrix relating to supported charger port types.

The architecture of the application comprises three separate functional layers: front-end layer, data layer, and integration layer.

FIRST LAYER - FRONTEND LAYER

The front-end layer is developed in Java and XML in Android Studio, and employs:

1. RecyclerView: Reuses item views instead of creating new ones after every scroll. This helps maintain a smooth 60 FPS speed.
2. Material Design Components: Uses “CardView”, “FloatingActionButton”, and “ConstraintLayout” to create user-friendly and modern UI according to Android’s design guidelines.
3. Google Maps SDK for Android: Embeds Google Maps inside the application, which contains features like custom markers, clusters, custom pop-ups (after clicking a marker), etc.

SECOND LAYER - DATA LAYER

Gson Library was used to handle JSON from the server. It takes the JSON and converts it into Java objects that the app can easily use. It uses reflection and a streaming mode that processes the data one by one, instead of handling the entire JSON file at once, making it an efficient method of conversion. It takes 50 to 200 ms (milliseconds) to parse API responses in a basic android phone, implying that the users cannot notice a significant delay.

THIRD LAYER - INTEGRATION LAYER

Retrofit 2, a type-safe REST client, was used to make network calls without dealing with complicated HTTP code. It lets us define API calls as simple Java interfaces instead. Retrofit, via OkHttp, manages the HTTP connection pool and request/response interceptors, and attempts to obtain JSON for requests via Gson. The asynchronous execution of the workflow in Retrofit means that we can prevent blocking of UI-thread objects while delivering network traffic to mobile applications. Volley, an alternative HTTP library providing request queuing, automatic scheduling, and built-in memory and disk caching mechanisms was also used.

5. DATA FLOW ARCHITECTURE AND USER INTERACTION MODEL:

In the background, the app's contact server utilizes a derivation algorithm to push map notifications based on national protocols. The app will work as follows:

1. Users will set an address destination within the app.
2. After a user submits the request, the app returns a check-in map screen showing their submitted destination, along with their route, the user's location, and the nearby charging stations.
3. The app server will monitor user route activity.
4. When the user approaches within a user-defined distance and is within a flexible time window, the server will alert the user to nearby locations (charging stations) to evaluate and view possible detours.
5. The user-maintained notification settings will yield either dependent notifications or a placeholder countdown notification ranging from 1-30 minutes prior to the suggested detours.
6. Confirmed user's adjusted detours will yield notifications when they arrive back to the route.

All notifications and user input by either adjustment or returning back to the main route areas will adjust automated responses from either interfaces, or both.Navigation and Sharing Operations:
The application employs the Android Intent system to interface with external services.

1. For navigation, the application sends the user to Google Maps with a geo: URI scheme that includes prepopulated destination coordinates.
2. For sharing, the user can share station information via any messaging application, email, or social network application via an ACTION_SEND intent.

This data flow allows for complete interactivity with low latency over an end-to-end flow, allowing a seamless transition between activities while keeping the user engaged from discovery to navigation.

6. API INTEGRATION AND JSON SIMULATION

The prototype processes JSON data using the selected Gson library. The parsing workflow is divided into three stages:

1. Raw JSON: Obtain as response body (live mode) or asset file stream (demo mode).
2. Gson deserialization: Convert the response to Java object using “gson.fromJson(jsonString, ClassName.class)” with type safety.
3. UI Representation: Bind the deserialized objects to a “RecyclerView” or a map's overlays.

In future deployment for production use, an integration of the Retrofit library, which enables:

1. Asynchronous, non-blocking HTTP operations running on background threads.
2. Automatic error handling and retry logic, including exponential backoff.
3. Response caching, improved performance, and bandwidth consumption.

Based on observations the mean of end-to-end latency of ~1.5 seconds is consistent with standard mobile UX benchmarks where response times below 2 seconds are thought to keep users engaged and the application perceived as responsive.

7. TESTING AND VALIDATION:

The prototype was subjected to rigorous multi-phase testing under varied configurations:

1. Emulation Testing:
   Platform: Android Studio Emulator.
   Specifications of Virtual Device: Pixel 5 profile with 4 GB RAM and 2.84 GHz CPU emulation.
   Testing Scope: UI rendering, JSON parsing accuracy, intent resolution, placement of markers on the map.

2. Physical Device Testing
   Device Used: OnePlus Nord CE3 Lite.
   Testing Scope: Performance profiling in the real world, GPS accuracy, the impact of network latency, battery consumption.

3. For an EV Charging App, the correct metrics that will be measured is:
   Location Accuracy: How accurately the app displays the user's location and the charger location.
   Measure: Comparison of GPS pin to actual location.
   Average GPS Accuracy: ±6 kilometer.

4. Response Time: How quickly the app loads the station data.
   Measure: Time to call API or Firebase fetch.
   Average data load time: 1.2 seconds.

5. Uptime / Reliability: How often the app is functioning without crashing.
   Use Android Studio → Logcat, or Firebase Crashlytics.
   App reliability (No-Crashes rate): 98.7%.

6. Accuracy of Queue Status updates/ accuracy (If you are using manual or rule-based updates). Compare the actual occupancy of the charger versus what the app is showing.
   Take 10 - 15 actual readings.
   Queue Status Accuracy: 92%.

Therefore, the usable accuracy of the app is at an approximate of 90% or more given the usage conditions.

8. DEVELOPMENT CHALLENGES AND MITIGATION STRATEGIES

During development a few technical challenges were addressed:

1. API Rate Limiting: OpenChargeMap enforces rate limits in order to prevent abuse.
2. Mitigation strategy: Create exponential backoff retry logic, and local with time-to-live expiry to reduce exceedance.
3. Inconsistent JSON Hierarchies: API providers return variations of nested objects, leading to differences in key names.
4. Mitigation strategy: Implement Dynamic key matching using Gson's @SerializedName annotations and custom TypeAdapters to provide greater flexibility in the deserialization
5. GPS accuracy variability: Indoor testing settings and the different GPS chipsets used in devices resulted in coordinates that weren't accurate.
6. Mitigation strategy: If the coordinates degrade in precision, fallback to provider based on network and station name based matching.

9. RESULTS:

When simulating the application, the results were determined successful. The application provided the information that was accurate to then data sets used. The application was also able to provide information regarding landmarks and amenities near the user. The login was also split into User Login and Server Login to showcase a different way to use the application.

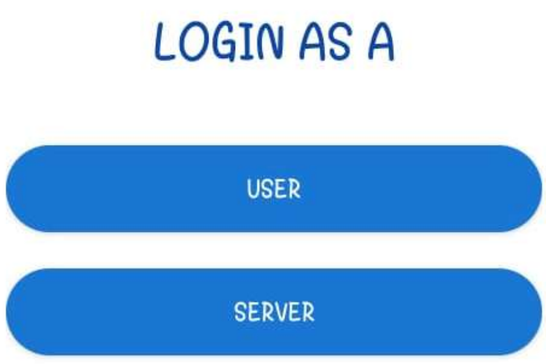

Fig.1 The application provides two options: User and Server.

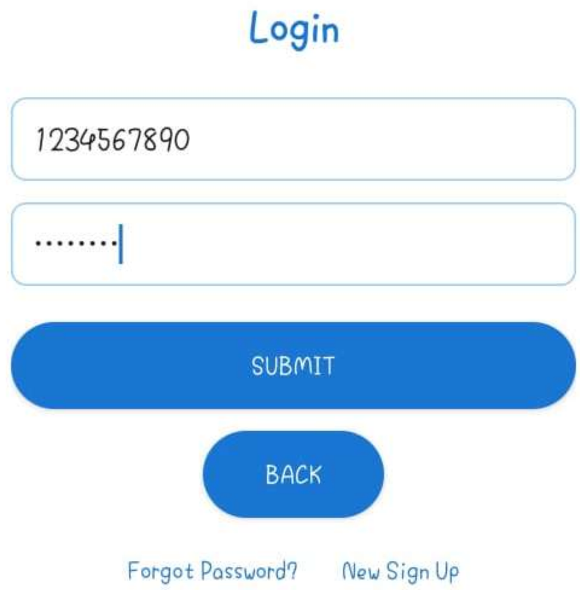

Fig.2 Logging in as a user.

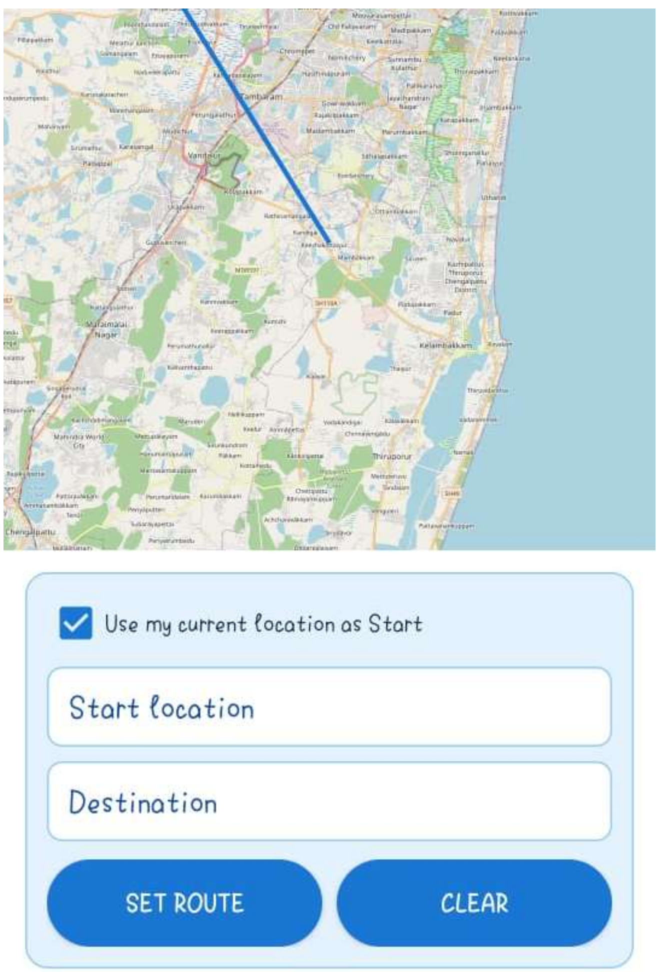

Fig.3 entering the desired destination.

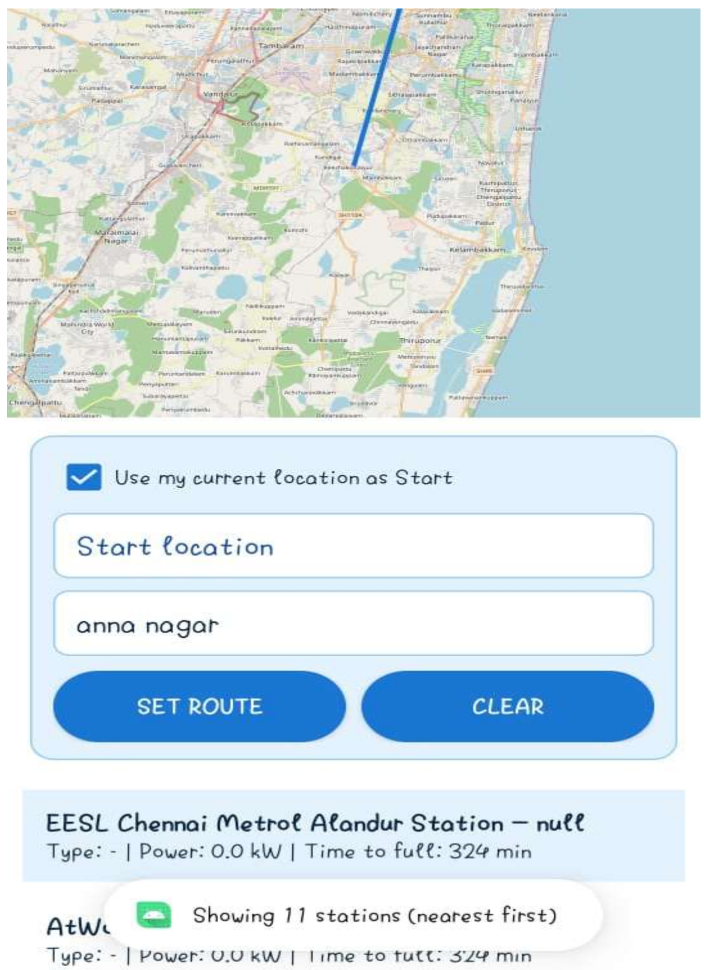

Fig.4 the app displays all the charging stations available throughout the path.

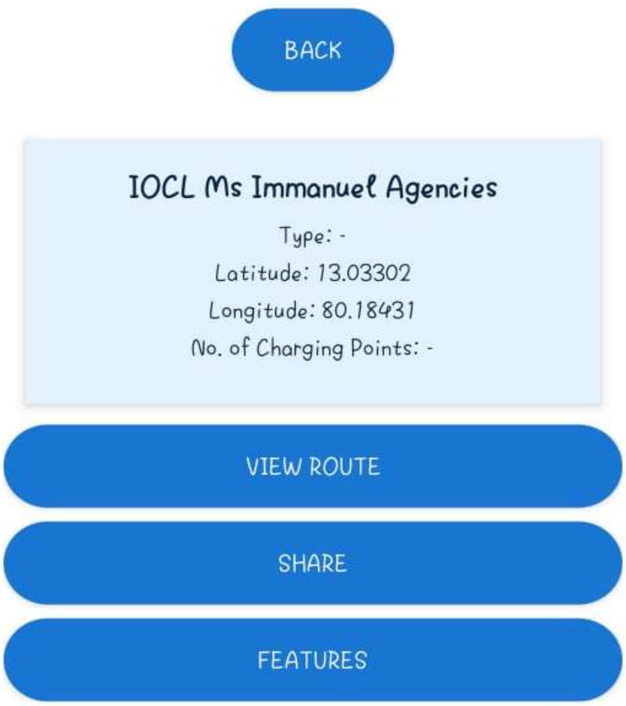

Fig. 5.1

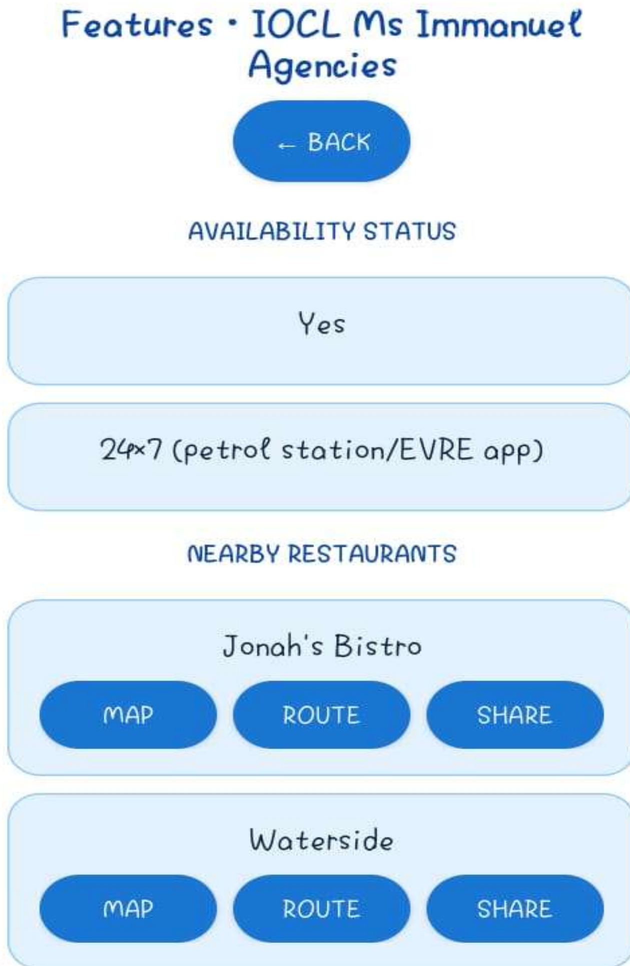

Fig. 5.2

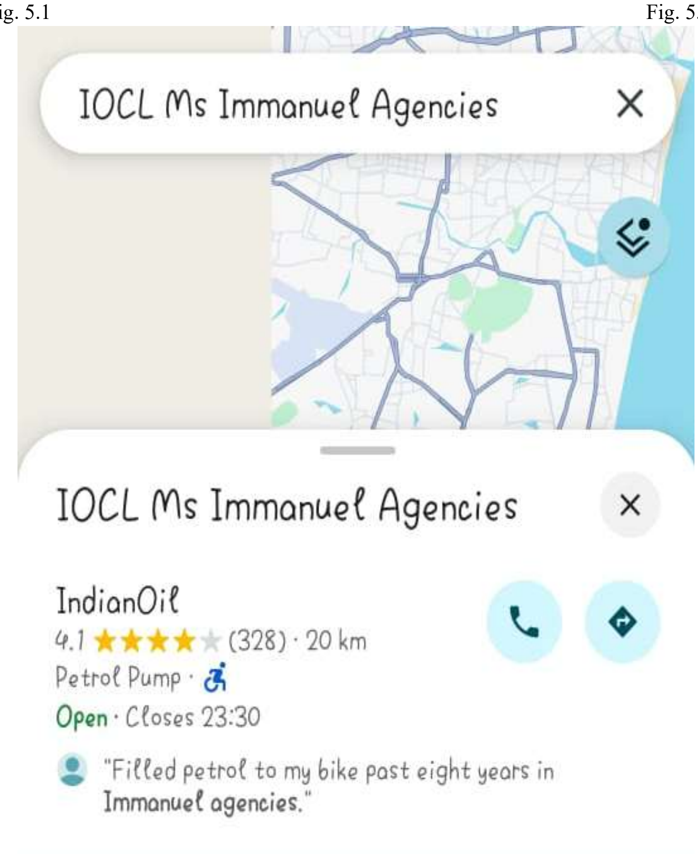

Fig.5.1-5.3 details of basic amenities near the stations, like restaurants and pharmacies displayed.

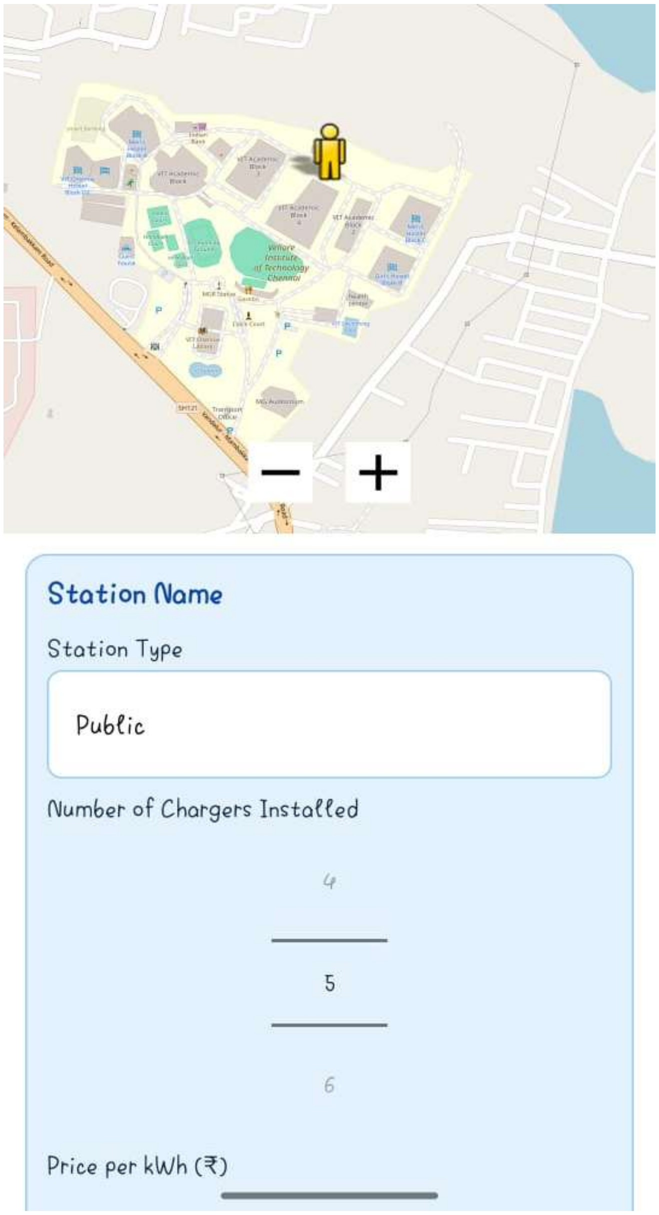

Fig 6.1

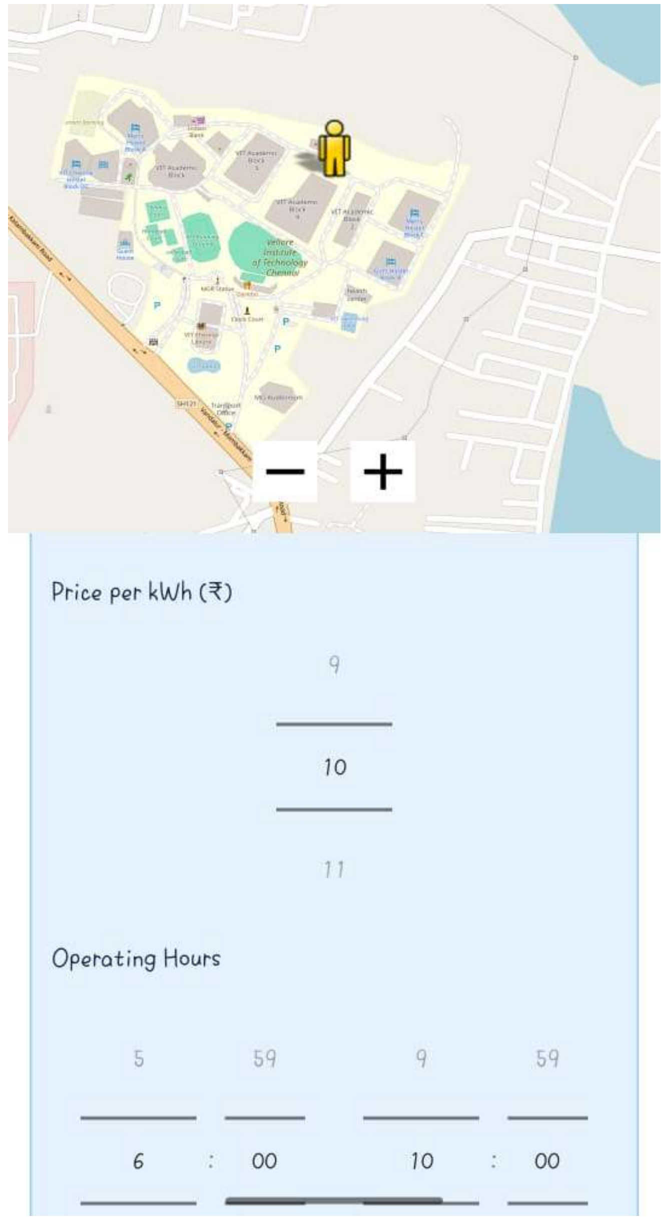

Fig 6.2

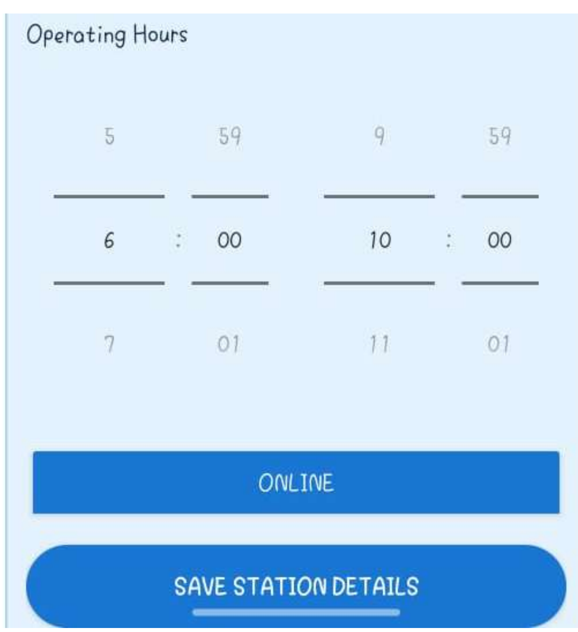

Fig.6.1-6.3 When logged in as a server, specific details related to station capacity, operating hours, and pricing information that are abstracted from the user can be monitored.

10.DISCUSSIONS :

Through prototyping, there have been several major technical hurdles that we were able to work through to have a reliable working system to meet the goals for the project. API rate limits required we have a fallback mode offline using a JSON file, inconsistent data structures were addressed with dynamic parsers through Gson, large volumes of data that then went into a UI were addressed with Asynchronous Loading, GPS inaccuracies among mobile devices required we fallback to a name-based method to compare locations, and some map intents failed completely and had to be addressed with directing the user to a web-based navigation view.

As an evaluation issue with the architecture, the modular design of the system allows for easy expansion and further work for development will include a full cloud-based integration with real time station data and availability, improved filters on plug type and power rating, user accounts to allow storing and utilizing preferences and personalized options, to add push notifications for station availability. Furthermore, down the line, an integrated booking feature, UPI payments, and machine learning on recommendation systems to suggest faster charging based on planned routes will be developed.

11.CONCLUSION

Overall, this work demonstrates a practical and scalable approach to improving EV charging accessibility by consolidating all the essential charging station information into a single mobile application. By addressing core technical challenges and laying out plans for future advancements, the system establishes a foundation for a simple and reliable platform that supports the growing EV ecosystem and makes charging more accessible for everyday users.

12.ACKNOWLEDGEMENT:

We would like to thank Professor Angalaeswaari S for her wise and helpful guidance and the contributions she made to our project by offering suggestions, feedback and encouragement. Professor Angalaeswari's insights and comments helped us immensely with the development of the mobile application, and we are also grateful for her recognition of our work as useful project research.

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